Report: Multimodal IMDB Analysis with Keras

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1. Introduction:

This assignment investigates the use of deep learning techniques for multimodal classification of films into genres using the IMDB dataset. The primary objective is to analyse how effectively two different models a Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM) network can classify films based on visual and textual data respectively.

The dataset consists of two modalities: film posters (image data) and film overviews (text data), with each film assigned multiple genres, making this a multi-label classification problem with 25 classes. This increases task complexity, as models must predict multiple relevant labels simultaneously. A CNN is used to classify posters by learning spatial features, while an LSTM processes overviews to capture semantic information, enabling a comparison of their performance and respective strengths across different data modalities.

2. Models & Methodology:

The CNN model was built using a custom ResidualBlock to create a deep network capable of learning hierarchical visual features from 64x64 poster images. The LSTM model was a sequential network using a pre-built vocabulary and a bi-directional LSTM layer to capture contextual information from film overview text. Both models were trained for multi-label classification across 25 genres, using binary cross-entropy loss. Model performance was monitored using a ModelCheckpoint callback to save the best weights based on validation loss.

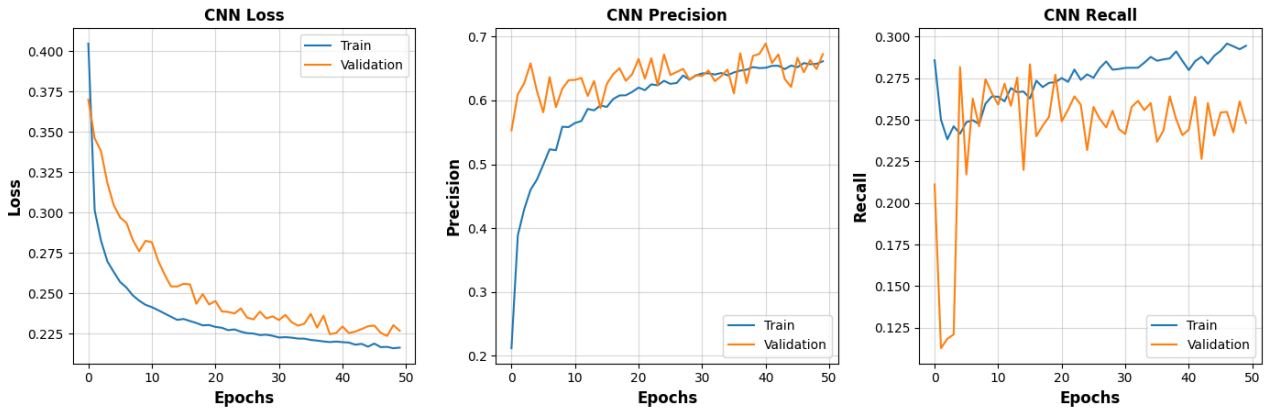
2.1. Image Processing: The image preprocessing pipeline standardises posters by resizing them to 64×64 pixels, converting them to float32, and normalising pixel values to [0,1], ensuring consistency and numerical stability. Data augmentation, including random horizontal flipping and brightness adjustments, is applied to improve generalisation while preserving semantic content.

2.2. Natural Language Processing: The text pipeline processes film overviews using batching (32), shuffling, and prefetching for efficiency. A TextVectorization layer converts text into integer sequences with a vocabulary size of 10,000, using output_mode='int'. The vocabulary is built using encoder.adapt() on the training data, ensuring that frequent and meaningful words are captured, providing an effective representation for the LSTM model.

3. Results and Critical Analysis:

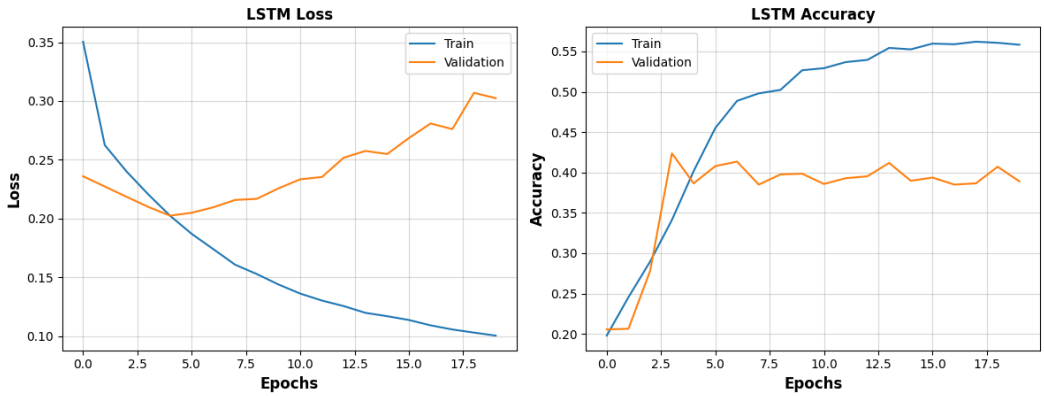
3.1. Quantitative Performance Analysis: The training history plots for both models reveal significant differences in their learning behaviour and generalization capability.

Figure 1: CNN Training Performance



Analysis of Figure 1 (CNN training curves): The CNN's training loss shows a steady, healthy decline, and the validation loss follows a similar downward trend, indicating good generalization without severe overfitting. However, the Precision and Recall metrics tell a different story. While training precision and recall improve consistently, their validation counterparts are much lower and fluctuate considerably. This divergence is a classic sign of overfitting for the specific task of multi-label classification.

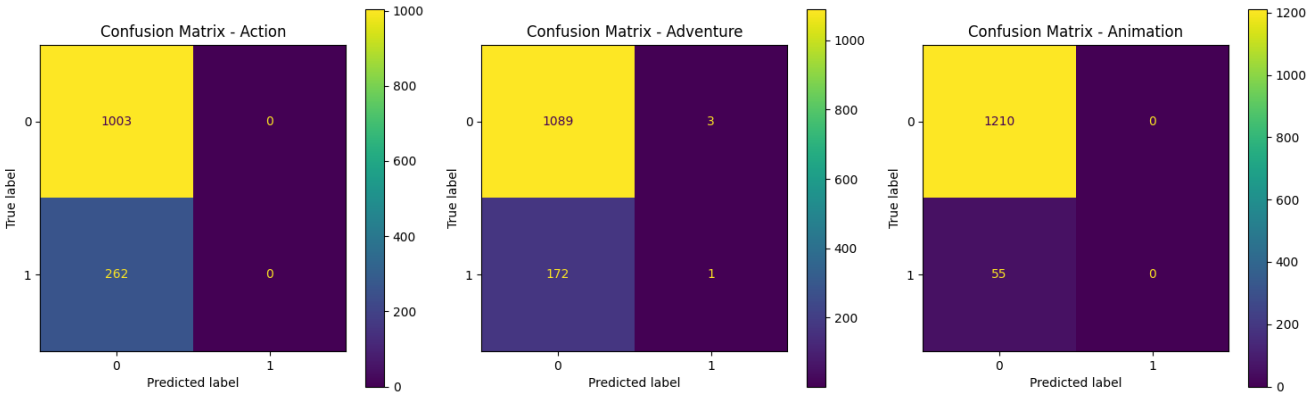
Figure 2: LSTM Training Performance



Analysis of Figure 2*:* The LSTM's performance is markedly different. While the training loss also decreases, the validation loss begins to increase sharply after epoch 4. This is a textbook example of overfitting. The model is memorizing the training overviews rather than learning generalizable linguistic patterns for genre classification. The validation accuracy plateaus and then slightly declines, confirming that beyond a certain point, further training is harmful. This suggests the model may be too complex for the size of the training data, or that the dropout rates (0.5 and 0.8) are insufficient to prevent co-adaptation of neurons.

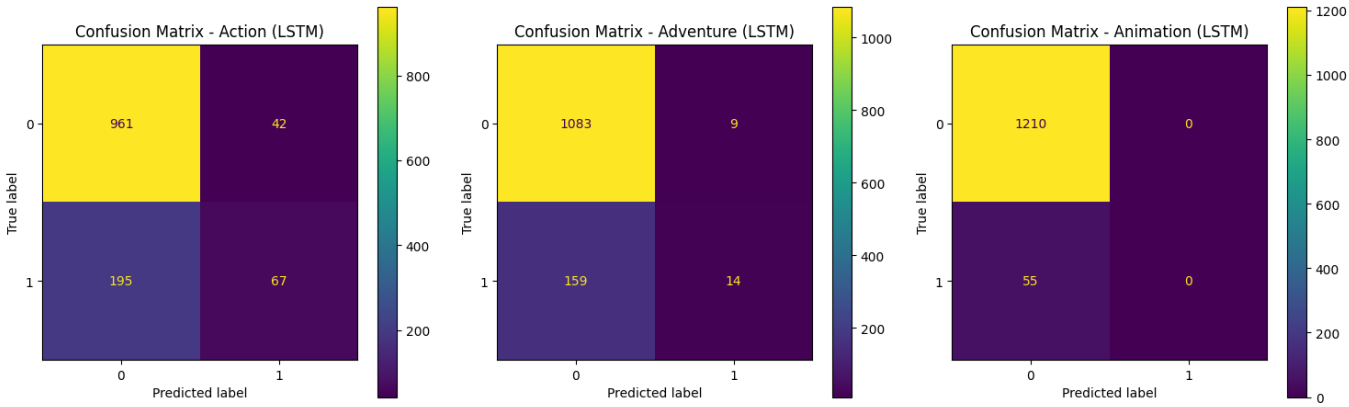
3.2. Confusion Matrix Analysis: Confusion matrices provide a clearer picture of the model's performance across individual classes by showing the distribution of true positives, false positives, and false negatives.

Figure 4: Confusion Matrices for Top 3 Genres (CNN)



Analysis of Figure 4: The CNN's confusion matrices for 'Action', 'Adventure', and 'Animation' show a high number of false negatives (the model failing to predict a genre that is present). This aligns with the low validation recall seen in Figure 1. The model is cautious, often predicting no genre rather than risking a false positive. This is likely influenced by class imbalance, where certain genres appear less frequently, making them harder for the model to learn. The model seems better at predicting the dominant visual theme but misses secondary or more subtle genre cues.

Figure 5: Confusion Matrices for Top 3 Genres (LSTM)



Analysis of Figure 5: The LSTM's confusion matrices show a different pattern, with a higher number of false positives. The model is overconfident, frequently predicting a genre that is not present. This overconfidence is another symptom of overfitting, as the model has learned to associate specific word combinations with a genre too rigidly, without understanding the context that would rule out that genre.

3.3. Qualitative Analysis: Case Studies: As shown in Figure 3, examining specific examples is crucial for understanding the practical strengths and weaknesses of each model.

- **Example 1 *(Drama – Angela's Ashes):*** Both models correctly identify *Drama* as the dominant genre. The CNN assigns high probability to Drama, likely due to the sober visual tone of the poster, while the LSTM also performs well by capturing narrative cues from the overview. This shows strong agreement between visual and textual signals.
- **Example 2 *(Mystery/Thriller – Lost Highway):*** The CNN struggles to identify the correct genres, predicting mostly Drama and Action, suggesting difficulty in interpreting abstract or stylistic posters. The LSTM performs better by identifying Comedy and Drama, but still misses Mystery/Thriller, highlighting limitations in capturing nuanced or implicit genre cues from text.
- **Example 3 (Action/Comedy/Romance – *Main Hoon Na*):** The CNN successfully captures multiple correct genres, including Drama, Comedy, and Romance, showing good multi-label capability from visual features. The LSTM predicts related but slightly misaligned genres (e.g., Crime instead of Action), indicating partial understanding of the narrative but less precise genre mapping.

Across these examples, several key patterns emerge:

- The CNN performs best when strong visual cues are present (e.g., *Main Hoon Na*) but struggles with abstract genres such as Comedy or Drama.
- The LSTM performs better when explicit keywords and narrative descriptions are available.

Figure 3: Comprehensive Example Analysis (CNN & LSTM)

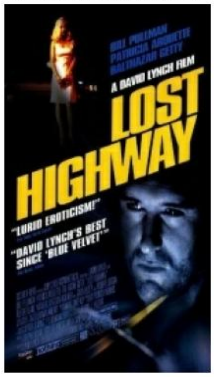
True Genres: Drama

Overview:
Based on the best-selling autobiography by Irish expatriate Frank McCourt, Angela's Ashes follows the experiences of young Frankie and his family as they try against all odds to escape the ...

CNN Top-3 Predictions:
- Drama: 0.49
- Comedy: 0.49
- Action: 0.38

LSTM Top-3 Predictions:
- Drama: 0.77
- Comedy: 0.33
- Romance: 0.22

True Genres: Mystery, Thriller

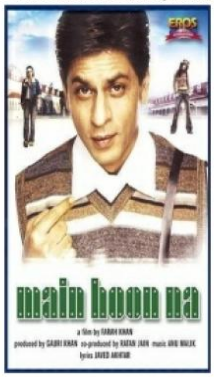


Overview:
After a bizarre encounter at a party with a stranger, a jazz saxophonist is framed for the murder of his wife and sent to prison, where he inexplicably morphs into a young mechanic, gets released, and begins leading a new life.

CNN Top-3 Predictions:
- Action: 0.44
- Drama: 0.43
- Comedy: 0.30

LSTM Top-3 Predictions:
- Comedy: 0.78
- Drama: 0.48
- Romance: 0.39

True Genres: Action, Comedy, Romance



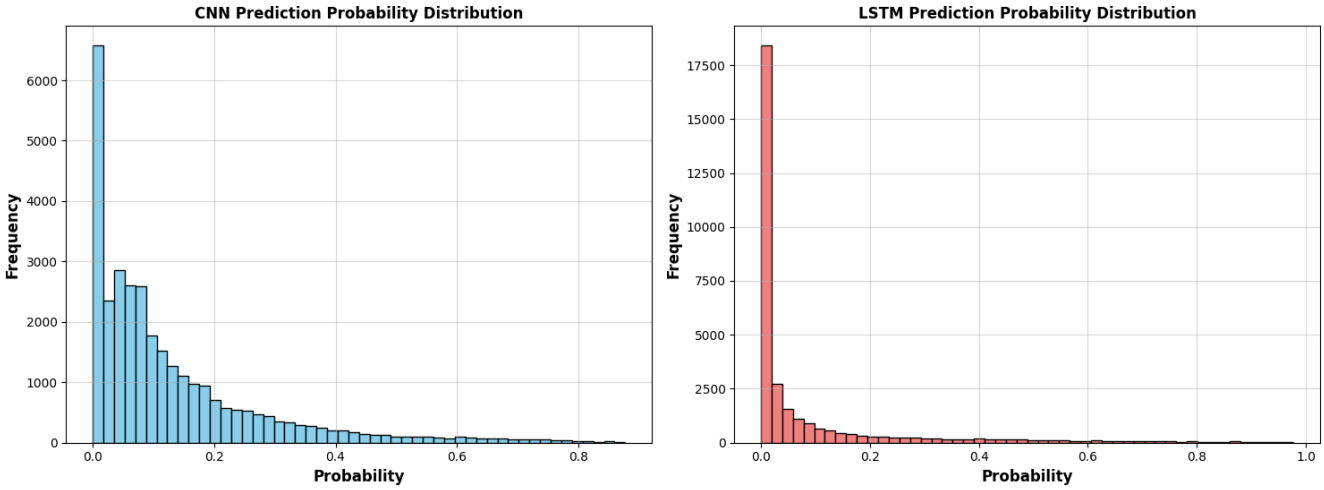
Overview:
An army major goes undercover as a college student. His mission is both professional and personal: to protect his general's daughter from a radical militant, and to find his estranged half-brother.

CNN Top-3 Predictions:
- Drama: 0.67
- Comedy: 0.50
- Romance: 0.44

LSTM Top-3 Predictions:
- Crime: 0.49
- Comedy: 0.48
- Drama: 0.45

3.4. Overall Model Comparison: The CNN and LSTM models operate in fundamentally different feature spaces as shown in Figure 6. The CNN extracts spatial features from images, while the LSTM captures semantic relationships from text.

Figure 6: Prediction Probability Distributions for CNN and LSTM



This suggests that visual features provide broader generalisation, while textual features offer higher specificity but lower robustness.

While the CNN demonstrates better generalization, this may be due to the relatively consistent nature of visual patterns in posters compared to the variability of natural language. However, textual data often contains explicit genre indicators, which can be highly informative when properly generalised.

The CNN performs well when strong visual cues are present, whereas the LSTM performs better when clear descriptive language is available. This suggests that both modalities provide complementary information, and neither model is sufficient in isolation.

3.5 Critical Evaluation and Limitations:

The results highlight clear differences in how the two models learn from their respective modalities. The CNN demonstrates more stable generalisation, as visual patterns in posters tend to be more consistent across samples. However, its lower recall indicates that it often fails to identify less visually dominant genres, suggesting a reliance on dominant visual cues rather than a full understanding of multi-label relationships.

In contrast, the LSTM model captures meaningful semantic information from text but suffers from overfitting, leading to reduced generalisation on unseen data. This is likely due to the variability and ambiguity of natural language, as well as the model's tendency to rely on prominent keywords rather than deeper contextual understanding.

A key limitation of this approach is that the two models are trained independently. As a result, they are unable to leverage complementary information from both modalities, which restricts overall performance particularly in complex multi-label scenarios where visual or textual cues alone are insufficient.

4. Conclusion:

This analysis reveals a fundamental trade-off. The CNN model, operating on visual poster data, shows better generalization but is hampered by low recall, often failing to identify genres that are not visually dominant. The LSTM model, operating on text, initially learns faster but quickly overfits, leading to a model that is overconfident and less reliable on unseen data. The superior generalization of the CNN suggests that, for this specific task and dataset, visual features provide a more stable and transferable signal for genre classification than the narrative text of overviews.

Future work could focus on improving the LSTM's generalization through stronger regularization (e.g., increased dropout, weight decay) or by leveraging pre-trained language models. The key insight remains that these modalities are complementary; a truly powerful system would fuse both visual and textual information, allowing a model to recognize both the visual style (from the poster) and the narrative themes (from the overview) to make a more informed and accurate prediction.